Interactive Sinhala Text Reader for the Visually Impaired Using OCR, NLP, AI.

R26-DS-002

Project Proposal Report

Sathsara S.P.Y B – IT22004468

B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Computer Science

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

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I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Sathsara S.P.Y.B	IT22004468	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Ms. Adya Dissanayake	19/3/2026	

ABSTRACT

Speech recognition technology and voice interaction is a vital technology for accessibility of visually impaired people in the digital world. There are many assistive applications that rely on speech recognition and voice interaction technologies in order to enable the visually impaired users to access the digital content by interacting with technology through speech and without depending on visual output. While there are many contributions in speech recognition for the Sinhala language and developing voice based applications for accessibility, current speech recognition and voice based applications are mostly based on simple speech to text functionality and command recognition functionality. They do not provide any conversational experience to the users and there is no personalization for individual users based on their preferences and interactions. Hence visually impaired users have very limited and non personalized user experience with current assistive technologies.

This research work aims to resolve this challenge by proposing an Adaptive Conversational Personalization and Voice Interaction Module for wise access through intelligent voice interaction. This proposed system allows the visually impaired users to communicate with an assistive platform by using intelligent voice communication using natural spoken language in Sinhala language. The proposed module will assist to access the system through voice communication. It will be able to recognize the voice input of the users, speech to text conversion, intent identification through applying Natural Language Processing (NLP) techniques, generation of personalized queries, communication with a central language model, and adapt the output response to the user preference models and finally deliver the output through text to speech technique using Sinhala language.

Speech Recognition, Intent Detection, Personalization, Large Language Model, Sinhala TTS will be used to develop the conversational voice interface to support accessibility. We aim to develop an “Intelligent Assistive Interaction Module” to provide visually impaired users with an opportunity to communicate using a natural and personalized way.

From the social perspective, this system can highly enhance the digital accessibility and autonomy of visually impaired people from the commercial perspective, this technology can be embedded into various assistive devices like smart glasses, accessibility applications, voice assisted information systems.

Keywords: **Conversational AI, Voice Interaction, Personalization, Assistive Technology, Speech Recognition**.

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Finally, I want to thank my fellow group members, who contributed to the structure and direction of the overall system. While the group collaboratively determined the overall scope and component outlines, each member independently invested to develop and refine their respective components. This ensured both a balanced distribution of work and a high degree of specialization for each component.

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LIST OF ABBREVIATIONS

Abbreviation	**Full Form**
AI	Artificial Intelligence
API	Application Programming Interface
ASR	Automatic Speech Recognition
AT	Assistive Technology
CCA	Connected Component Analysis
CNN	Convolutional Neural Network
CS	Conversational Systems
DB	Database
DSR	Design Science Research
HCI	Human Computer Interaction
LLM	Large Language Model

MAT	Medial Axis Transform
ML	Machine Learning
NLP	Natural Language Processing
OCR	Optical Character Recognition
SBAS	Speech Based Accessibility Systems
SDK	Software Development Kit
STT	Speech to Text
TTS	Text to Speech
VI	Visually Impaired
VUI	Voice User Interface
WER	Word Error Rate

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1. INTRODUCTION

1.1 Background

Text is very essential for education, employment and everyday life. Visually impaired people face a large number of difficulties in dealing with printed newspapers, textbooks, notice boards and other types of information documents. Utilization of assistive technologies such as screen readers and Optical Character Recognition (OCR) software has eased the reading of text in many languages, but it is still very difficult for the visually impaired to read the texts in Sinhala language. They face immense difficulties because Sinhala language support is not provided sufficiently and reading Sinhala scripts is a complex task.

Based on the breakthroughs of Artificial Intelligence, Computer Vision and Natural Language Processing, there is now the possibility to create intelligent reading systems capable of transforming printed texts into speech. The technology at the base of these applications works as follows: an image of a page of a book is taken and, through an Optical Character Recognition (OCR) technology, the text of the page is identified and can then be translated into audio through a technology called text-to-speech (TTS). Currently, such technologies are applied to simple purposes: for instance, the purpose of readers for the blind is to make possible the reading of printed texts. Currently, such readers are able to recognize and read the text on a page but they often do not offer an interactive level of communication, other than the simple outputting of the recognized text as speech.

People with visual impairment interact with technology using their voices. So it is very important that these technologies can work in the best possible way with voice interactions. Today, most of the assistive reading tools that were developed to facilitate access to text for the visually impaired, allow only for one way communication, they only convey information by reading out the text they have identified. The user has no chance to query the application with respect to the read information, or to ask for clarification on some item. They can also not intervene during the reading process and choose alternative ways of presenting the content.

Whereas speech interactions are typically carried out in conversation form (with give and take of information), and most speech-based assistive systems (Audio Based Assistants) lack conversational properties in the first place, there are other aspects that one might naturally wish a speech ABA to personalize for a particular user. These may include such attributes as speech pace, amount of description provided, format of speech output for lists, output for single items and so forth. (For example, one user might prefer summaries versus full descriptions; another user that has limited sight and decreased acuity might wish speech to be delivered at a slower rate, with announcements provided for repetitive statements about the output provided by the speech system and also with simpler versions of the content available.) A typical speech ABA, however, generally produces the same speech output in a given circumstance without regard for differences from one user to another.

Today society faces a number of challenges. One of them is the provision of skills to meet social needs. We attempt to introduce a solution to meet the needs of the time carrying out

research on the Adaptive Conversational Personalization and Voice Interaction Module for an intelligent Sinhala assistive reading system for visually impaired and low-vision people. This module will help people to communicate to an assistive reading system by voice using human to human conversation using voice inputs and give human to human conversation output using voice based on the user preference and past interactions. The proposed system will combine speech recognition, intent identification, conversation management and personalized speech generation in order to provide a more dynamic and personalized experience for reading to visually impaired and low-vision people.

The Adaptive Voice Interaction and Personalization Engine is the main communication interface between the user and the IRC (Intelligent Reading System). Unlike a typical TTS (Text-To-Speech) engine, our module allows for voice conversation, understands user intentions, captures user preferences and provides auditory responses that are meaningful and relevant to each user. Our goal is to transform traditional assistive reading technologies from passive tools to highly interactive and personalized systems, which can greatly benefit visually impaired and elderly low-vision users.

1.1.1 Problem Statement

The assistive technologies (ATs) are developing at a very high pace and the current reading systems for visually impaired users (VI) have many limitations. The current electronic reading systems mainly depend on the text recognition technique for conversion of text into speech. These systems do not facilitate interactive conversation with the content and the visually impaired user has to passively listen to the speech generated from the recognized text. Hence, the system has limitations in terms of user interaction and personalization of the content.

Traditional assistive voice interfaces usually follow a speech-to-text and text-to-speech flow. The speech-to-text converts speech to input, and the text-to-speech technology does the output conversion. However, the speech-to-text functionality is only capable of recognizing what is said without being able to recognize the user's intent, taking the context into consideration, or adapting to user behaviors. Therefore, there is very little interaction between the user and the interface.

Currently, there are number of voice-based assistive systems being proposed and developed for VIs and OLVC. However, as highlighted earlier, these systems have many shortcomings. One of the major shortcomings is the lack of personalization for the users. VIs may have different levels of familiarity with technology and they require different types of information for different situations. Similarly, OLVC may have different levels of understanding and knowledge with respect to information they require in different situations. Unfortunately, current assistive voice-based systems do not take into account differences among users. They deliver the same output to all users, regardless of their familiarity with technology and the nature of information they require and the way they interact with the system.

Many current reading technology systems fail to support conversational exploration of document content. For example, users are not able to ask follow-up questions to clarify or seek additional information about specific details in a document, ask for summaries of individual document sections, or clarify statements or examples that they do not fully understand. As a result, these systems do not fully support the flexible, conversational processes that are often required for understanding lengthy or complex content.

An intelligent voice interaction system providing adaptive conversational communication and personalized information services is highly desired for the visually impaired. The Adaptive Conversational Personalization and Voice Interaction Module targets this required functionality by employing user intent identification, interaction behavior analysis, and personalized output generation. Through a dynamic and user-centric voice interaction approach, the proposed module can increase the accessibility of reading for visually impaired and low-vision users.

1.2 Literature Review

Nowadays Speech Recognition, Conversational AI and Intelligent Voice Interfaces technologies are trending. We can build voice based applications and our interaction with digital applications will be by speech or voice. Voice interaction is highly useful to the visually handicapped. Though there are many studies done to recognize speech in Sinhala and to design Voice Based Systems, they are more focused on speech recognition, and commanding applications. None of them designed conversational models that can interact with users in a human manner. These conversational models have to learn about users behaviors and preferences to respond accordingly.

1.2.1 Sinhala Speech Recognition Research

Speech recognition technology is a key technology that allows the user to communicate with the assistive system using speech. Many research works are carried out to develop an Automatic Speech Recognition (ASR) system for the Sinhala language.

Weerasinghe (2021) has developed an enhanced version of the speech recognition system for the Sinhala language using time-delay deep neural networks architecture along with the Kaldi toolkit, which yields a word error rate of about 25.12% [1] The study shows that deep neural networks can yield a speech recognition system with greater accuracy when dealing with the Sinhala language, especially for speech recognition tasks. The study mainly focuses on improving speech recognition accuracy and does not deal with the conversational aspects or the response generation for an interactive chatting system.

In our earlier work, Wijesekara and Ranathunga investigated the feasibility of using end-to-end deep learning models for speech recognition of Sinhala language, and showed that the modern deep learning techniques can capture the required acoustic variations for the particular speech characteristics of the Sinhala language[2]. Most of the research studies

carried out so far focuses on technological aspects of the speech recognition for a particular language. None of these studies has focused on facilitating human conversation using speech recognition technology.

In Speech Recognition, another work carried out by Wickramasinghe et al. explores the possibility of developing a real time subtitle generator for the Sinhala language. They employed MFCC features along with speech recognition models obtained from CMUSphinx for the development of this application [3]. These findings depict that speech recognition can be a reliable tool for speech to text applications where language is Sinhala. Nonetheless, this speech recognition system could only facilitate speech to text functionality and has no scope for establishing conversations, or adapting speech responses based on user preferences.

He proposed a system for Speech recognition of Sinhala to be used for IVR applications using Hidden Markov Models [4]. The proposed system will be able to identify pre-defined spoken commands and numbers in telephone applications. Karunanayake's speech recognition system for telecommunication services though a pioneering effort in Sinhala speech recognition is limited to pre-defined words and therefore it does not involve free conversation and learning the vocabulary.

This research shows that it is possible to develop a speech recognition system that can understand Sinhala speech. While a considerable amount of work has been done to improve the accuracy of speech recognition and command recognition, not much work has been done to develop a conversational voice user interface that can have a two way communication with the user.

1.2.2 Conversational Voice Interfaces and Dialogue Systems

Conversational systems (CSs) allow users to interact with applications via conversations instead of using command languages. Dialogue systems have recently been studied to gain insights into how users intend to interact with CSs and how CSs can effectively respond to user dialogues.

The first paper "Intent Recognition Framework for Low Resource Spoken Dialogue Systems" by Gupta et al. suggests an intent recognition framework that aims to improve intent classification for low resource languages and hence, achieves a significant increase in accuracy for limited training data [5]. Intent recognition is a crucial element in conversational voice interfaces.

In our previous work, Borgeaud et al. explored the use of language-agnostic embeddings for intent classification. They showed that training models on multilingual data using representation learning can significantly improve the performance on intent classification tasks for multiple languages [6]. Due to the lack of large amount of training data available for low resource languages such as Sinhala, our primary interest is multilingual representation learning for intent classification.

A significant body of work in the literature has explored approaches to identifying user intent and emotions via analysis of language input. While these investigations are valuable for advancing the field of dialogue systems, very little has been done in terms of fully realizing the development of a complete conversational interface, capable of appropriately responding to user feedback and preferences over time.

In addition, Hoy (2018) makes mention of the rise of voice assistants like Google Assistant, Amazon Alexa and Apple Siri, which can engage with users using speech [7]. Obviously, these types of conversational interfaces are relevant to accessibility for people with visual impairments. However, these commercial products are generally geared towards a broader range of applications other than documents and do not offer the specialized adaptation required for VIs.

1.2.3 Personalization in Voice Interaction Systems

Personalization is a central issue for intelligent user interfaces, and especially for accessibility technologies where users often have vastly different requirements. We will see how adaptive user interfaces can make use of user behavior and preferences to adapt their interactions to meet specific needs of individual users.

A User-Adaptive System is, as defined by Fischer in 2001, “A system that automatically adapts or presents its behavior on the basis of observations about the characteristics of, or the actions or interactions of, one or several users, or on the basis of a model of what such users or groups might be” [8]. This type of system enhances the usability of its user interface by customizing its behavior on the basis of specific user information.

More recent studies in human–computer interaction include adaptive conversational systems where dialogue responses are adapted to user preferences, past dialogue interactions, and contextual factors [9]. Providing more human-like interaction between humans and computers is a prime goal of human–computer interaction research.

Even though speech recognition, understanding and generation technologies have been improving over the years, current commercial speech-based accessibility systems, which we refer to as Speech Based Accessibility Systems (SBAS), provide users with a set of static speech-based outputs, without learning their behavior. As speech is often the sole mean of interaction for visually impaired people, having a system with unpersonalized static speech-based outputs leads to a highly non-productive Voice User Interface.

1.2.4 Limitations in Existing Research

From the reviewed literature, several key limitations can be identified:

• The existing works in the field of Speech Recognition in Sinhala language are mostly focusing on improving the accuracy of the Speech Recognition system rather than improving user experience towards human-to-human conversation.

• Existing systems typically support command-based voice input rather than natural dialogue.

• Current dialogue systems emphasize intent detection but lack adaptive personalization mechanisms.

• Research on adaptive conversational interfaces tailored for visually impaired individuals in Sinhala speaking context is scarce.

All the mentioned limitations make the need for an intelligent system highly felt, which integrates speech recognition, intent detection and the generation of responses accordingly, in order to enable personalized conversational interaction for visually impaired and low vision users.

This activity aims to address the research gap by developing the Adaptive Conversational Personalization and Voice Interaction Module (ACP-VIM) that supports real-time user interaction through Voice User Interface (VUI) and personalize its responses based on the users preferences and behavior.

1.3 Research Gap

While speech recognition and conversational technologies have advanced in many ways in recent years, they still present numerous challenges and opportunities in speech-assisted interaction particularly for visually impaired (VI) users, especially in low-resource scenarios such as speech interaction in Sinhala language. The existing speech recognition systems as well as state-of-the-art Voice User Interfaces (VUIs) do not support dynamic speech adaptation, which poses considerable challenges for many visually impaired people who prefer to use speech to interact with technology and are particularly hampered in low resource settings like in Sri Lanka.

All the previous work on speech technologies carried out on the Sinhala language concentrated on the construction of the Speech recognition system by applying various machine learning and deep learning techniques. Thus, they were only concentrated on obtaining maximum accuracy for speech recognition which is considered to be the first step to constructing an interactive conversation based dialogue system to facilitate user's interaction with systems such as information search etc. As the proposed systems do not consist of more than speech recognition, users can only have a basic interaction as text is not produced using output speech synthesis for the user input speeches which were correctly identified using the Speech recognition system.

There are many voice based applications developed for the Sinhala language which heavily depend on command based interaction model where the user has to use pre-defined commands to get a response from the system. This does not really support natural conversation flow. The users have to remember all the commands to be able to use these

systems efficiently. Which also reduces the usability of the voice based assistive technologies for visually impaired users who solely rely on the voice interfaces.

Current dialogue systems and intent recognition models primarily rely on spoken input to detect user intent, and do not always consider response adaptation to user preferences, dialogue history or context. Without personalization, the resulting explanations provided to users may not be relevant to their individual needs, which can be a significant challenge for users with varying levels of visual impairment or comfort with technology.

There are also various other limitations such as majority of conversational systems work on widely used languages and commercial purposes. In contrast, there has been hardly any research done on the adaptive conversational interfaces that are applicable for a conversational interface that is tailored for Sinhala speaking visually impaired users. Hence, there is no appropriate solution available for providing a personalized voice-based interaction within the Sinhala assistive reading systems that we proposed earlier.

Therefore there is a need for a system that can advance speech recognition and command-based interaction beyond the current basic level. The system should be able to carry out an adaptive conversational communication and personalize the responses. The Adaptive Conversational Personalization and Voice Interaction Module that we propose will attempt to fill this gap by incorporating the speech recognition, user intent detection and interaction-based personalization mechanisms, in order to advance the voice interaction experience for visually impaired and low-vision users.

2. OBJECTIVE

2.1 Main Objective

The proposed outcome aims to develop the Adaptive Conversational Personalization and Voice Interaction Module that will allow visually impaired and low vision users to communicate with assistive reading system for visually impaired using a conversational interface through voice. This will be achieved by conducting activities such as speech recognition, intent identification and generating contextual responses to user inputs.

2.1.1 SMART Objectives

- Specific: - The goal of this objective is to develop the Adaptative Conversational Personalization and Voice Interaction Module for voice interaction for visually impaired and low vision users.
- Measurable: - This work proposes a language understanding module using an ensemble strategy for spoken language interface applications, and assesses the efficacy of the proposed language understanding module according to the speech recognition accuracy, intent recognition accuracy, response relevance and user satisfaction in voice interaction.
- Achievable: - This goal is thought to be achievable by leveraging established speech recognition and natural language processing technologies including intent classification models and text-to-speech synthesis, and in a manner consistent with the goals of this Research Initiative.
- Relevant: - The objective contributes to the overall research goal of improving accessible information interaction for visually impaired users via adaptive voice-based communication.
- Time-Bound: - Research activities will be completed through the design, implementation and evaluation of research findings within the agreed timeframe of the project, and tangible outcome measures will be provided by the end of the project.

2.2 Specific Objectives

1. **To develop a Sinhala speech recognition interface** that converts user voice commands into textual input for system processing.
2. **To design and implement an intent detection mechanism** capable of identifying user requests from spoken input to support conversational interaction.
3. **To develop a user interaction analysis mechanism** that records user preferences and interaction patterns to enable adaptive personalization.

4. **To implement a personalized response generation mechanism** that adjusts explanations and information delivery based on user preferences and interaction history.
5. **To generate adaptive Sinhala speech responses using text-to-speech technology** that improves accessibility and clarity for visually impaired and low-vision users.
6. **To evaluate the performance of the adaptive conversational voice module** using defined evaluation metrics such as interaction accuracy, response relevance, and usability.

3. METHODOLOGY

3.1 Research Approach

This research is implemented by applying Design Science Research (DSR) approach which aims at designing and evaluating artifacts to solve real-world problems. The artefact designed and evaluated in this study is Adaptive Conversational Personalization and Voice Interaction Module to enhance the voice interaction experiences for visually impaired and low-vision individuals.

To investigate the conversational interface design for assistive reading systems, the proposed research aims at developing a voice interaction module that can understand the spoken words in Sinhala language, identifies the user intent and provides the relevant response by applying the conversational adaptation strategies. The proposed system will be implemented and tested to verify whether the adaptive personalization will enhance the voice interaction usability and the usefulness in assistive reading systems.

This method allows us to work towards our goal of creating a functional system component, while also providing us with performance metrics for evaluating our overall system performance.

3.2 Technical Methods and Algorithms

The proposed Adaptive Conversational Personalization and Voice Interaction Module leverages multiple artificial intelligence and natural language processing technologies to facilitate conversational interaction and adaptive/relevant response generation.

User voice input will be recorded and converted as text using Speech-to-Text (STT) model that can handle Sinhala speech. After converting speech to text, the intent of the user request should be identified. Intent can be identified using machine learning approaches or using transformer language models.

After determining the user intent, the Personalization layer will attempt to understand the user preferences and their past behavior. It will keep a lightweight User Model or User Profile that contains the following information’s preferred output format, frequency of responses, amount of output provided. The Personalization layer will then use this information to tailor the output proportionate to what is understood from the user's behavior and preferences and to adjust the output structure and length.

Process the output and send it to the TTS engine to produce the audio output in Sinhala language. In this use case the speech output is adjustable, for example users can vary the speech tempo and pitch and this can be highly useful for the visually impaired and elderly low vision users.

Speech Recognition, Intent Detection and Response Generation will be integrated in this module in order to enable a more human to human interaction between the user and the assistive reading system.

3.3 Tools and Platforms with Justification

Component	**Tools / Platforms**	**Justification**
Speech Recognition (STT)	Whisper / Sinhala ASR models	These open source software tools are built on a set of deep learning architectures that can process low data amounts, provide high levels of accuracy when compared to traditional rule-based systems and provide increased accuracy for Speech Recognition (IR) of Sinhala..
Intent Detection	Python NLP libraries (Transformers, BERT-based models)	Models developed using Transformer architectures provide excellent capability in classifying user Intentions and performing natural language understanding tasks, making them an effective approach for identifying user intent in Conversational Systems.
Personalization Engine	Python, Machine Learning libraries (Scikit-learn, PyTorch)	ML frameworks based on Python allow for creating models that are capable of adapting to user interaction patterns enabling the generation of personalized responses.
Voice Output Generation (TTS)	Sinhala Text-to-Speech APIs / Google TTS	TTS technologies make it possible to provide clear understandability of Sinhala language output through voice for understanding visually impaired users who cannot read.
System Integration	Python-based backend environment	Python is a highly versatile language that is very widely used in developing machine learning (ML), artificial intelligence (AI) and natural language processing (NLP) technologies. It can also be very easily integrated with other software packages providing support for the development and deployment of the other components of the overall system (Voice Recognition, Intent Identification, and Personalization Engine)

Table 1 - Tools and Platforms

3.4 Data Collection and Ethical Considerations

This feature requires Speech Interaction Data and User Interaction. For the development and evaluation of this feature, the Speech interaction data and User interaction data have to be provided. The training and testing of the speech interaction functionality such as speech recognition and intent identification will have to be done with Sinhala speech samples. The speech and intent identification datasets may include, open sourced Sinhala speech datasets as well as speech samples recorded by using the system for testing purposes.

Data collected by this service will be used for personalization purposes such as selecting recommended products to show, for recognizing spoken search queries, for analyzing frequency of interaction with certain elements and for analyzing user preferences. The data will be treated according to the principles of scientific research.

Information regarding this study will be provided to all participants involved in system testing and their consent will be obtained prior to data collection. All personal data will be deleted and users' data will be anonymized.

3.5 Validation Strategy and Performance Metrics

The evaluation of the Adaptive Conversational Personalization and Voice Interaction Module will be conducted through both the analysis of technical performance metrics and the assessment of the user experience.

Speech recognition performance will be evaluated based on the WER and accuracy. The intent detection model will be evaluated based on standard classification metrics including precision, recall and F1-score.

The Personalization mechanism will be tested with User Interaction Tests using visually impaired and low vision participants to validate its functionality. The results of the tests will provide information regarding the relevance of the response to the user action, as well as the success rate of the interactions, and user satisfaction.

The metrics will be used as criteria for the research to verify whether the proposed adaptive conversational module improves the functionality and usability of the voice interaction in the assistive reading systems.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of The Proposed Solution and Its Components

This proposed system is an Intelligent Sinhala Assistive Reading Platform that allows users with visual impairments, as well as those with low vision, to easily obtain printed information using a voice interactive interface. Using the capabilities of several technologies (computer vision; natural language processing (NLP); and conversational-based artificial intelligence technology), this system captures printed Sinhala text, intelligently processes those texts; and provides the user with adaptive voice responses.

There are four primary functional components that make up the entire system architecture. These are defined as follows, with each functional component representing a different functional stage in the overall processing pipeline.

Component 1 - Topological Sinhala Character Reader

Captures and corrects the input image, isolates headline text using thresholding and CCA, skeletonizes characters via MAT, and classifies them through a CNN model and a Structural Recognition Model, before applying rule-based correction to produce the final headline text output.

Component 2 - Context-Aware Post-OCR Error Correction

Extracts the standard body text using a baseline OCR and pre-processes and corrects the OCR's output via a fine-tuned Transformers model built using synthetic noisy-clean groups of Sinhala text and finally produces a contextually corrected Sinhala text output.

Component 3 - Advanced Document Retrieval and Information Generation

Retrieves content from local databases, websites, and previous scans and generates vector embeddings representing the Semantic Content for Semantic Search and provides the retrieved context to an LLM with Cross-Session Memory to generate either simplified or expanded responses.

Component 4 — Adaptive Conversational Personalization and Voice Interaction

Captures the user's spoken input and detects its intention(s) using a Mini-LLM, producing a tailored voice response based on what was learned from the user's preferences and by conducting a pattern analysis of the user's previous interaction behaviors, and delivers the voice response to the user using Text-to-Speech (TTS) technologies.

4.2 System Diagram of Components

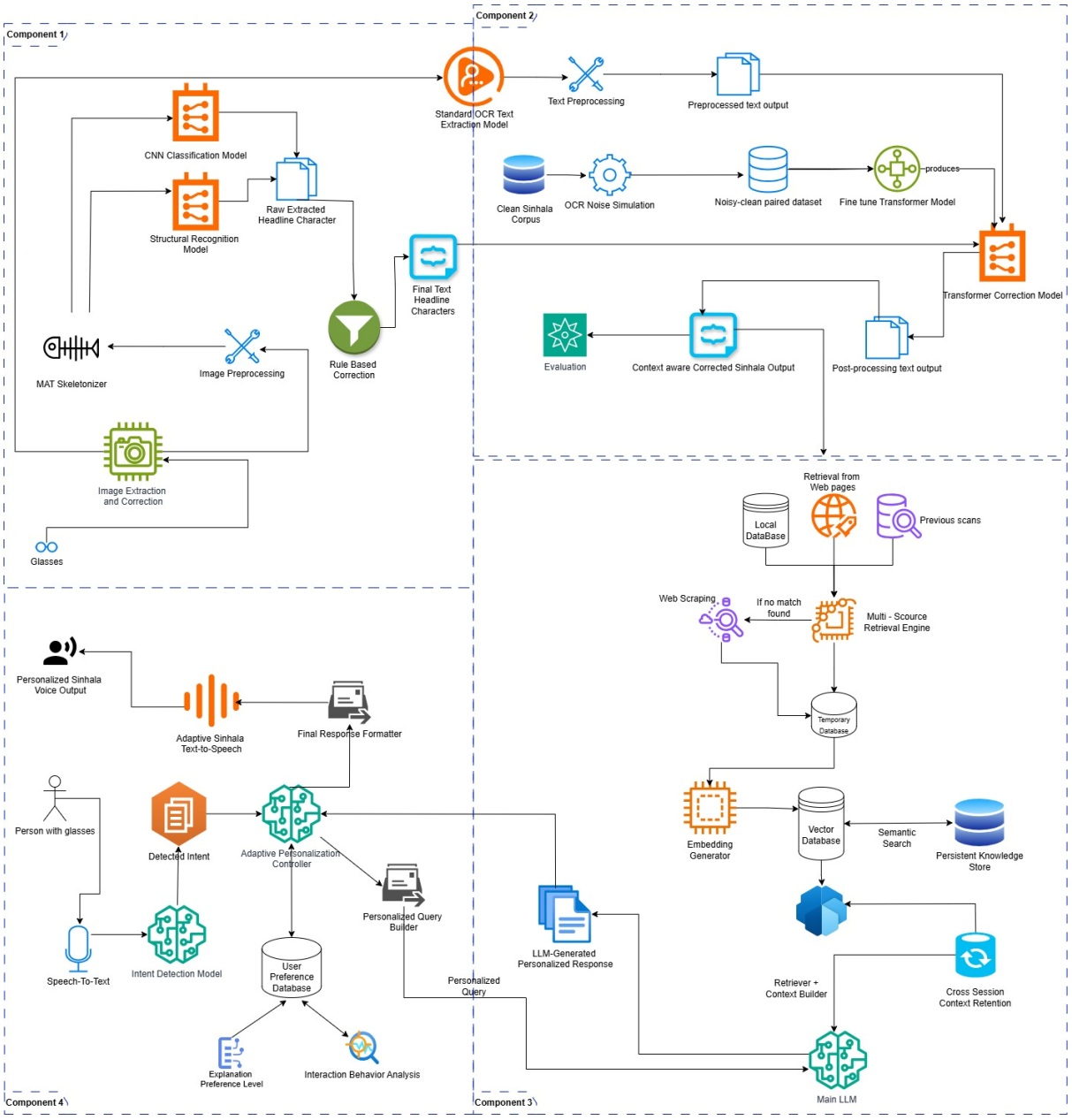


Figure 1 - High-Level System Architecture of the Sinhala Assistive Reading System

This diagram presents the complete system architecture and illustrates how the four major components interact with each other to process printed Sinhala content and generate personalized voice responses for the user.

4.3 Highlight of the Individual Component and Its Parts

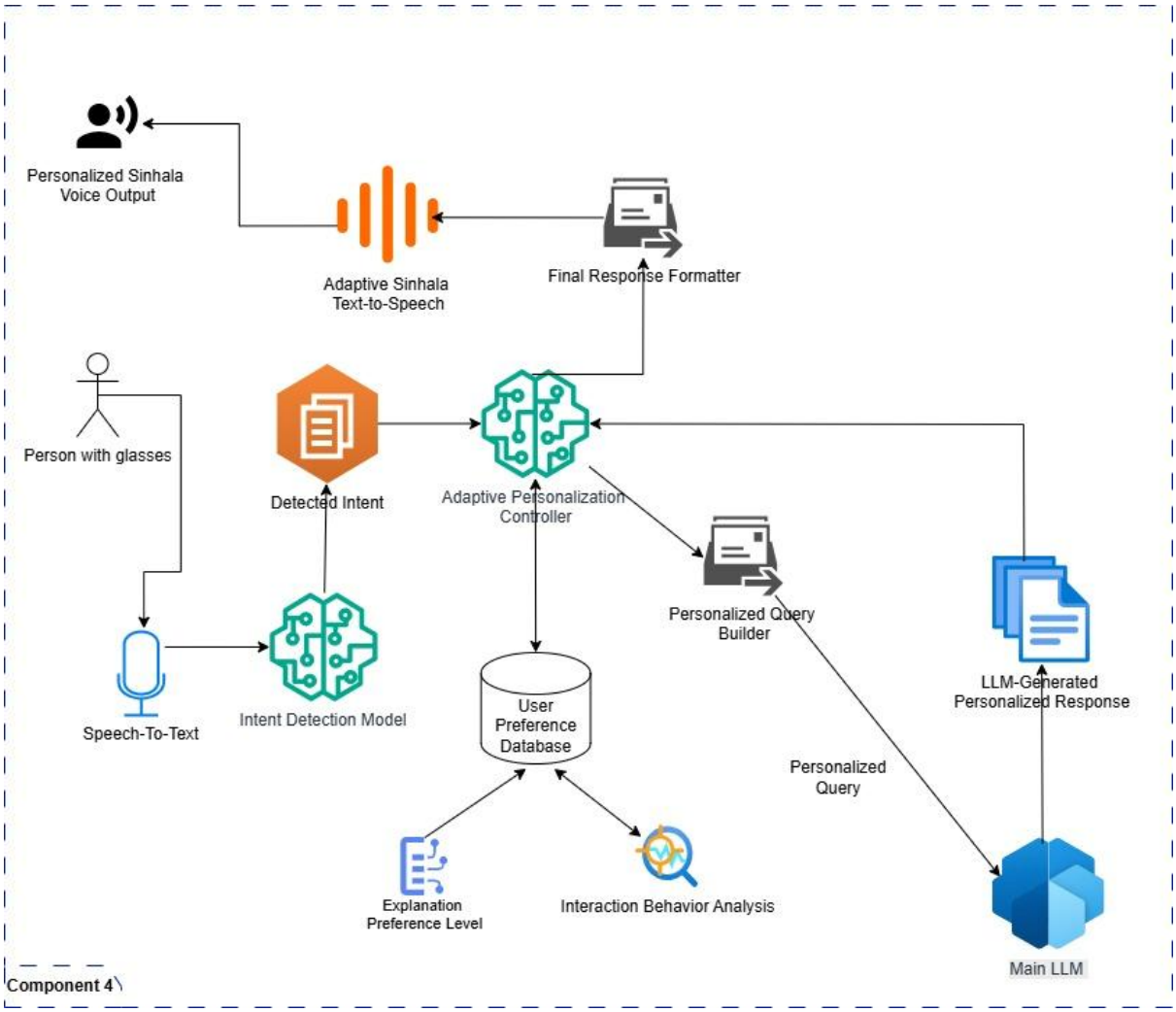


Figure 2 - Architecture of the Adaptive Conversational Personalization and Voice Interaction Module

The Adaptive Conversational Personalization Module is composed of several sub modules that are responsible for enabling conversational dialogue flow and generating contextualized human responses.

Speech-to-Text (STT) User will convey the request/command to the system by speaking to it. Speech to Text module will be used to translate the spoken input in to machine understandable text in Sinhala language.

Intent Detection Model The output of this module is given as input to the intent detection model, which identifies which intent the user has expressed. The intents identify whether the user is asking to perform an action such as summarizing, explaining, rephrasing, or simplifying a document.

Adaptive Personalization Controller The intent is sent to the Adaptive Personalization Controller, a decision-making module. The Adaptive Personalization Controller uses the intent and user preferences and search history to generate a query that is personalized to the user.

User Preference Database The User DB database stores such things as users' explanation preference level, their desired level of response detail, and so on. Each record in User DB corresponds to a particular user logged in at a particular time.

Interaction Behavior Analysis This module examines past behavior of users and as a result, identifies a pattern of behavior which are saved and used to adjust and evolve the personalization parameters.

Personalized Query Builder Based on the detected intent and personalization parameters, a personalized query is generated and sent to the main language model in Component 3.

Final Response Formatter The output of the LLM is then routed to the response formatting layer, which formats it into a structure that can be read by the speech synthesizer.

Adaptive Sinhala Text-to-Speech (TTS) Finally, the formatted response is converted into a Sinhala speech that can be read by a visually impaired user using an accessible speech output interface.

4.4 Component Interactions

The system components interact with each other in a sequence in order to process the information captured from the document and to provide the relevant responses to the user.

Component 1 does the initial work of identifying Sinhala characters in images and it sends the identified text to the next Component 2. There in Component 2, with the aid of context-aware transformer models the OCR errors are being corrected.

The output of the correction algorithm is sent back to Component 3 and, after acquiring relevant data from various information sources, the information is explained through a language model.

Component 4 receives the Voice command and by understanding the intent of the voice command, and then applying appropriate personalization rules, constructs a query that is then sent to the LLM in Component 3. The output from the LLM in Component 3 is then taken by Component 4 for further adaptation and is then output as adaptive speech in Sinhala.

This interaction structure is implemented to achieve a robust document understanding functionality and an appropriate conversational dialogue interface.

4.5 Data Flow Explanation

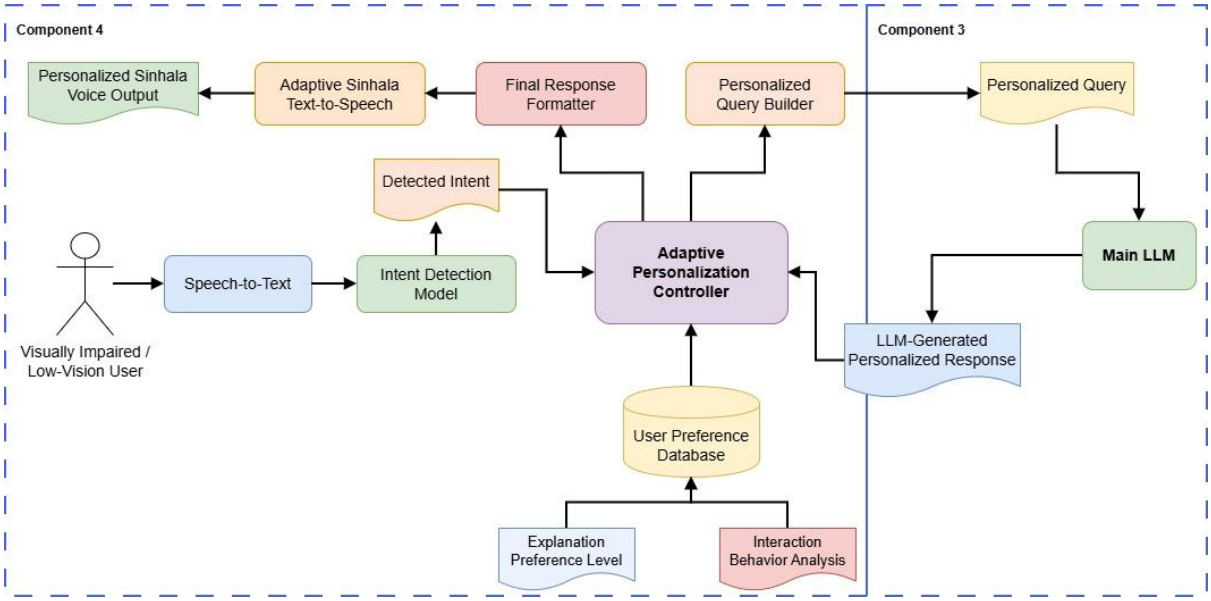


Figure 3 - Data Flow Diagram of the Adaptive Conversational Personalization and Voice Interaction Module

This stage receives the voice input of the user, transforms it to text using speech recognition and then the intent detection identifies the intent behind this request. The personalization controller then processes the user preferences and historical interaction behaviors to further refine this request.

And finally, the Personalized query is given to the Main LM to get the answer and after post processing and Voice generation by using the TTS module, the answer is given as speech output in Sinhala.

This pipeline enables our system to take a printed book and turn it into an audiobook personalized for each user.

4.6 Scalability and Security Considerations

Our system design is optimized for scalable and secure real-world deployments.

At an application level, the whole system is scalable at several levels. Firstly, the different subsystems within the speech-based interface such as speech recognition, personalization and language processing can be easily and dynamically upgraded. Other levels of scalability involve the use of the cloud and scalable machine learning algorithms for a large number of users and interactions. And, at the level of the individual application, the knowledge bases for the interfaces as well as the services they access can be dynamically modified. In addition, some individual subsystems and functionality can be updated or added as new modules.

The User Profile data stored in the user preference database is a security and privacy risk. When analyzed in conjunction with other data such as usage logs, information about user

interaction can be derived. We therefore need to ensure this data and the derived information is treated with care so as to not to impact the privacy of the users. We need to also ensure any personal data we get from users is removed or anonymized. Further, we need to implement appropriate authentication and storage methods for the data.

In particular, our design considerations take into account the potential impact on Assistive Technology (AT) environments, focusing on reliability, scalability and security.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

This page describes the Adaptive Conversational Personalization and Voice Interaction Module in terms of the users for whom it is designed. However, as will be seen, other stakeholders can also make use of, or benefit from, this module in a variety of ways. The primary users for whom this module is intended are people with visual impairments, such as blind and low vision, people looking for alternative ways to interact through digital technologies.

Primary Stakeholder

Visually Impaired / Low-Vision Users

Users who cannot see the content on websites or web applications and hence heavily depend on speech. Our system enables users to query, get explanations or information required through personalized voice in Sinhala. This helps to adapt the speech behavior of users and learn the interaction patterns and preferences, so that it can come up with more relevant responses over time.

Secondary Stakeholders

Elderly Users with Limited Vision

Virtual Assistant that helps seniors read smaller fonts & navigate through technology and get explanations in simpler language. The Elderly would benefit from using Virtual Assistant that can perform voice interactions and also provide an explanation of what the system does in simpler terms, which would help them to understand the functionality of Virtual Assistant more effectively.

System Developers and Researchers

The Developer will be responsible for the implementation, maintenance and development of personalization algorithms, voice interaction models, and the integration with the language model and other system components.

Assistive Technology Providers

Accessibility technology vendors have a wide range of options to deploy the features of the Software Development Kit (SDK) into their own products such as Assistive Applications for Smart Glasses or Mobile Accessibility Apps.

5.2 Functional Requirements

The following functional requirements describe the functional units of work that the Adaptive Voice Interaction and Personalization Engine should be able to perform.

1. The system shall capture the voice from the user and shall convert the voice to the Text using Speech-to-Text module.
2. The system shall utilize the intent detection model to ascertain the user's intent from the output text generated by the language translator.
3. The system shall retrieve user-specific interaction preferences from the User Preference Database.
4. The system shall analyze user interaction behavior to update personalization parameters.
5. The system should be able to construct a tailored query for each user based on the recognized intent and the user's preferences.
6. The system shall pass on the personalized search query to the main Knowledge Base LB (Language Model).
7. The system gets the output generated by the language model and applies it to the user requirements.
8. The system shall output the response in a format that is proportionate to the expected conversational response and explanation appropriate to the age of the user.
9. The system shall convert the final response into Sinhala voice output using a Text-to-Speech module.
10. The system shall provide real time speech output of the generated voices.

5.3 Non-Functional Requirements

Non-functional requirements describe the quality attributes a system must have to be reliable and efficient.

Usability The system shall include voice user interface technology to enable visually impaired users to interact with the system without the need to visually navigate the system.

Performance This needs to be a relatively low latency system to enable what we feel is a really conversational voice user experience.

Accessibility The system must be able to provide a fully voice-based user interface. One of the intended uses is to allow people who cannot (cannot learn to) use other interfaces to make use of it.

Scalability Handling the concurrent access of multiple users and maintaining the data of users for personalization is a key requirement.

Security User interaction data and preference information in the User Preference Database must be properly secured and inaccessible to unauthorized users.

Reliability The system should always give the same answer and maintain stable voice interaction performance during an ongoing conversation.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

Our primary target market includes the visually impaired and low vision persons and organizations that require easy access to information on the web.

Primary Market Segments: -

Visually Impaired and Low-Vision Individuals This category includes visually impaired users who can't access digital information. A technology system that will make use of speech interaction and provide adequate descriptions and other necessary information about the contents it presents will make their access to information easier.

Educational Institutions Braille Displays enable users to read on a reflectivity surface, similar to braille, via dot shifting technology. This makes Braille Displays ideal for schools and universities that have students that are blind or have low vision and require access to a variety of materials within the classroom.

Assistive Technology Organizations Accessible Technology Vendors We have established partnerships with companies that provide accessible technology products such as smart glasses, screen readers, and other accessibility applications. We can integrate our system with these products.

Public Information Services Speech Recognition technology is also widely used in Government services, libraries and public information portals to deliver Citizens Services to the visually impaired.

Customer Persona

- Name: Nimal Perera
- Age: 42
- Occupation: Teacher with Visual Impairment
- Needs: Access to information through voice interaction without relying on visual interfaces.

- Challenges: Difficulty reading digital content and navigating traditional web interfaces.
- Expected Solution: He will be able to ask and answer the questions required with the help of the voice and get the correct answer in Sinhala voice according to his ability level.

6.2 Value Proposition

In the proposed system a Personalized Conversational Assistant is implemented to help visually impaired people to communicate with information systems by using voice communication.

The key value offered by the system includes:

- Personalized voice responses tailored to the user's understanding level.
- Adaptive explanations based on user interaction behavior.
- Sinhala language support for improved accessibility.
- Integration with assistive technologies such as smart glasses.
- Improved digital accessibility for visually impaired individuals.

The system uses voice, personalization and artificial intelligence to enable humans to communicate with objects and services in more human and natural ways than is currently possible with today’s user interfaces that are based on screens and keyboards.

6.3. Revenue Model

The commercialization of the system can follow different revenue models depending on the implementation scenario.

Software Licensing Model Accessibility service providers and educational establishments buy a commercial license in order to embed our solution in their applications or platforms.

Subscription-Based Model Users or organizations are able to pay either monthly or annually for a subscription to the service to gain access to a tailored intelligent assistant.

API Integration Model Developers and companies can access the system through APIs and integrate it into assistive technology, apps, or platforms.

Partnership Model We have started collaborations with companies who develop assistive technologies as well as with organizations focused on accessibility with the aim of integrating our system to specialized products designed for accessibility.

6.4 Cost Estimation

Cost Category	Description	Estimated Cost
Development Cost	AI model development, personalization system, and voice interaction modules	Medium
Cloud Infrastructure	Hosting language models, databases, and API services	Medium
Data Storage	Storage for user preference data and interaction history	Low
Maintenance and Updates	System improvements, bug fixes, and model updates	Medium
Integration Cost	Integration with external systems and assistive devices	Medium

Table 2 - Cost Estimation

6.5 Intellectual Property Considerations

In regard to this system, there may be several potential areas of IP protection.

Software Architecture The design of the adaptive personalization/voice interaction system may qualify for software copyright protection.

Personalization Algorithm Algorithms associated with the adaptive personalization and behavioral analysis of users may qualify as proprietary technology.

AI Model Integration Framework The integration framework for intent detection, personalization, and language model interaction could also be classified as a unique system design and therefore may be eligible for protection.

Moreover, it is important that licensing and data protection policies exist to ensure compliance with accessibility and data privacy.

7. BUDGET AND JUSTIFICATION

Category	Item	Estimated Cost	Justification
Hardware (External)	Camera, Microphone	LKR 7,000 upwards	These are essential to build the system and even if we use pre-shot images, the better camera directly improves the project results.
Software	All tools open-source (Python, OpenCV, scikit-image)	-None	All primary tools are freely available
Data	.ttf files, Print media from website, capture images	-Depends, as documents to capture images can be bought or borrowed.	Ground truth dataset creation
Man-Hour	Student research time	-Not applicable	Already accounted for in the project
Hardware (Internal)	Computer	-Not applicable	Current available laptops will most be able to the software.

8. WORK BREAKDOWN STRUCTURE (WBS)

Task	Description	Deliverable	Duration (Weeks)
Requirement Analysis	Identify user requirements for voice interaction and personalization features for visually impaired users.	Requirement specification document	3 Week
Literature Review	Study existing conversational AI systems, accessibility tools, and personalization techniques.	Literature review summary	2 Week
System Design	Design the architecture of the Adaptive Personalization and Voice Interaction module.	Component architecture diagram	2 Week
Dataset Preparation	Collect and prepare voice interaction datasets and user preference data for system development.	Processed dataset for training	4 Week
Intent Detection Model Development	Develop and configure the intent detection model for recognizing user queries.	Trained intent detection model	3 Week
Personalization Controller Development	Implement the Adaptive Personalization Controller to manage user preferences and response adaptation.	Personalization engine implementation	5 Week
User Preference Database Implementation	Design and implement the database structure to store user interaction data and personalization parameters.	User preference database schema	2 Week
Personalized Query Generation	Develop the personalized query builder to integrate user preferences with detected intent.	Query generation module	2 Week
Integration with Main LLM	Connect the personalization module with the Main LLM	Integrated system communication module	4 Week

	component for response generation.		
Response Formatting and TTS Integration	Implement the final response formatting and adaptive text-to-speech generation.	Voice output generation module	3 Week
System Testing and Validation	Conduct functional testing, usability testing, and performance evaluation of the module.	Testing report	1 Week
System Optimization	Improve response accuracy, latency, and personalization performance.	Optimized system version	2 Week
Documentation and Final Report	Prepare final research documentation, system description, and evaluation results.	Final research report	1 Week

Table 3 - WORK BREAKDOWN

9. GANTT CHART

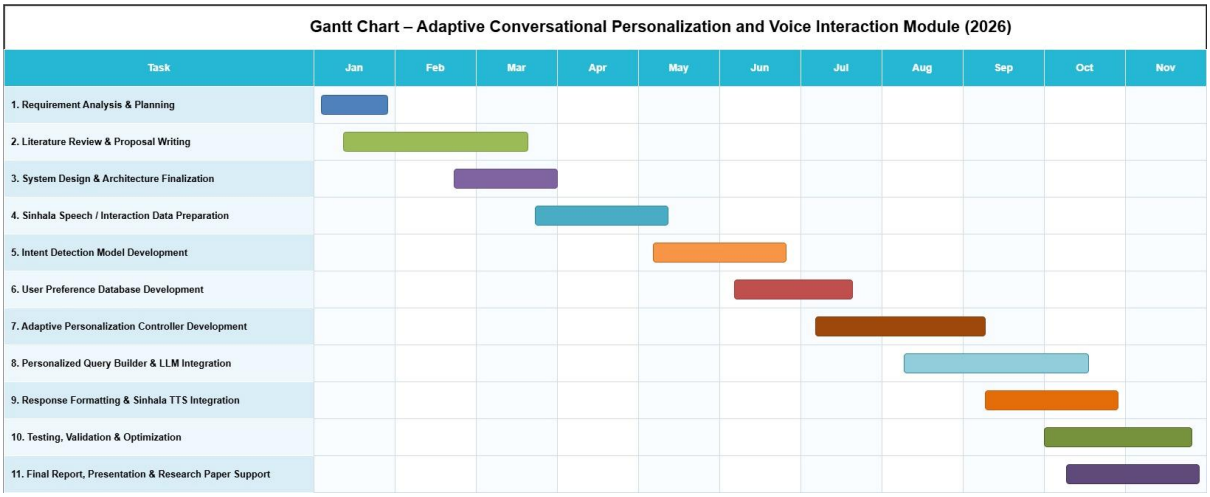


Figure 4 - Gantt Chart – Adaptive Conversational Personalization and Voice Interaction Module

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11. APPENDICES

Appendix 1 - Survey Instruments

11.1 Survey Instruments

A User Requirement Survey for Voice Interaction of Visually Impaired and Low Vision Individuals is proposed to investigate the interaction requirements of visually impaired and low vision users while interacting with voice based assistive systems. This survey aims at investigating user expectations, preferred interaction modalities and accessibility requirements.

The questionnaire includes the following key areas:

- User familiarity with voice-based technologies
- Preferred language and speech interaction styles
- Level of explanation required for system responses
- Challenges faced when using existing assistive technologies
- Expectations for adaptive conversational systems

These will be used to train our personalization model and iterate on our conversation design.

Appendix 2 - Detailed System Diagrams

11.2 Detailed System Diagrams

This appendix includes the technical diagrams that describe the architecture and workflow of the proposed system and the individual component.

The following diagrams are included:

- Overall System Architecture Diagram
- Adaptive Voice Interaction and Personalization Module Architecture
- Component Interaction Diagram
- Data Flow Diagram of the Personalization Engine

These diagrams illustrate how the voice interaction module communicates with the speech recognition system, the personalization controller, the user preference database, and the main language model.

Appendix 3 - Preliminary UI Mockups

11.3 Preliminary UI Mockups

Although the system primarily uses voice interaction, basic user interface elements may be required for system configuration and user preference management.

The preliminary mockups illustrate:

- User preference configuration interface
- Voice interaction status display
- Accessibility control options

These UI designs help visualize how users or administrators may interact with the personalization system during configuration or monitoring.

Appendix 4 - Risk Analysis

11.4 Risk Analysis

The development of the adaptive conversational voice interaction system may involve several technical and operational risks.

Risk	Description	Mitigation Strategy
Speech recognition errors	Sinhala speech recognition accuracy may affect system understanding.	Use improved ASR models and implement error correction techniques.
Limited training data	Sinhala datasets for conversational AI are limited.	Use multilingual models and transfer learning techniques.
Personalization accuracy	Incorrect personalization may lead to irrelevant responses.	Continuously update user preference models using interaction feedback.
System latency	Processing speech and AI responses may introduce delays.	Optimize system pipeline and use efficient inference models.

Table 4 - Risk Analysis

Appendix 5 - Ethical Considerations as Well as Data Protection

11.5 Ethical Considerations as Well as Data Protection

The research project involves collection of user interaction data and data processing. As a result, ethical considerations in addition to ensuring protection of user data, must also be addressed.

The following key principles regarding ethical conduct will also apply to this proposed research.

- User Consent - Participants will provide informed consent prior to data being collected about them in their user studies.

- Data Privacy - Both personal data as well as data regarding user interactions/logs will be stored in a secure manner and will only be used for research analysis after removal of identifiable information.
- Accessibility - The system will be designed to allow for visually impaired individuals equal access to the information provided by the system.
- Responsible AI Use - AI Models will be used for development in a transparent and responsible manner with respect to maintaining user privacy.

All collected information will strictly only be used for research purposes and will be in accordance with the institution's research ethics.